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Preference-Based Assessments

A Head-On Ordinal Comparison of the Composite Time Trade-Off and the Better-Than-Dead Method

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ABSTRACT

Objectives: The valuation of health states worse than dead is challenging. Currently used time trade-off methods face problems in (1) detecting time-dependent preferences and (2) insensitivity toward severity for states worse than dead. The better-than-dead (BTD) method has the potential to detect time-dependent preferences. This study compares the BTD and composite time trade-off (cTTO) methods at the ordinal level.

Methods: An experiment was conducted in a convenience sample in which respondents (N = 200) valued the same set of 7 health states in the BTD method and cTTO method. Binary BTD responses were used, with response categories of better than dead and worse than dead. Ternary cTTO responses were used, with the additional equal-to-dead response category. Polychoric correlations were used to determine the agreement between these methods. Consistency and test–retest reliability were assessed within methods.

Results: Overall agreement between the cTTO and BTD method equaled 77.1% and differed between health states and respondents. For both methods, there were few inconsistencies, and the test–retest reliability was comparable (88%). Health states were more often considered worse than dead in the BTD method (BTD: 54.7%, cTTO: 37.2%).

Conclusions: The high agreement between both methods and the comparable amount of inconsistencies and test–retest reliability suggest that the methods have similar measurement properties. The BTD method yielded higher frequencies of worse-than-dead responses while essentially asking respondents to make similar choices. This accounts for part of the disagreement between the methods. Several explanations are offered for this difference, yet more research is needed to explain this phenomenon.

Keywords: better-than-dead method, composite time trade-off, health utility assessment.

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Introduction

Health utilities are commonly used in economic evaluations in healthcare. Generic preference-based instruments such as the EuroQol 5-Dimensional (EQ-5D) instrument allow for the comparison of different health states on a single scale, ranging from 1 (full health) and 0 (dead), whereas worse-than-dead health states are assigned negative values. To measure health utilities, valuation methods such as the standard gamble, time trade-off (TTO), composite time trade-off (cTTO), and discrete choice experiments are used.[1-3]

Valuation methods face 3 important problems. The first is that there are difficulties in eliciting values for states worse than dead.[4,5] The second is that time-dependent values for health states cannot be detected by these methods.[6,7] The third is that insensitivity toward severity for worse-than-dead health states is observed in some methods.[8] To deal with the challenges in calculating values for worse-than-dead health states, the EQ-VT protocol was introduced. The EQ-VT is a standardized protocol for EQ-5D valuation studies that combines cTTO and discrete choice experiments, with the purpose of generating value sets for the EQ-5D instrument.[9,10] Another recently developed method is the better-than-dead (BTD) method, which leads to consistent weights for health state attributes and is able to detect time-dependent preferences for health states.[7] The BTD method produces values on the 0 (dead) and 1(full health) utility scale, in which the proportion of "dead" responses is used to infer a value for a health state. States that have a probability of 50% of being chosen over dead are then assigned the value 0, as the probability of picking dead or that health state is equal. This anchors the scale on dead equaling 0.

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It is unclear whether the cTTO and the BTD methods measure the same underlying preferences and have the same measurement properties. The goal of this study is to provide a head-on ordinal comparison of the cTTO and the BTD method and to assess a number of measurement properties of both methods. If both methods measure the same underlying preferences, agreement on whether a state is worse than dead or better than dead should be high within respondents. Second, inconsistencies in responses should also be comparable. Lastly, test–retest reliability should be comparable. Our research question is the following: what is the agreement between the cTTO and BTD methods, and how do these methods compare on inconsistencies and test–retest reliability? An experiment is set up to answer this research question, in which respondents complete both BTD and cTTO tasks for the same health states.

Methods

General Approach

The BTD method provides binary responses, that is, whether a health state is considered either better or worse than dead. For comparison's sake, we will look at ternary responses for the cTTO method. These ternary responses show whether a health state is considered better than, equal to, or worse than dead, as explained below. Ternary cTTO responses are compared with the binary BTD responses.

Data Collection

Data were collected from respondents in Nijmegen, the Netherlands. The target sample size was set at 200 respondents. A convenience sample of predominantly biomedical students and researchers at Radboud University Medical Center was used for this methodological study. Initially, respondents were given a €5 gift card for a popular web store as a reward for participating. This was increased to a €10 gift card later during data collection, to accelerate the inclusion of respondents. All interviews were conducted by the same interviewer (B.R.). The interviewer was trained in a half-day course at the EuroQol Research Foundation in Rotterdam and performed several practice interviews with colleagues before starting the data collection.

Health State Classification System

The health state classification system used in this experiment is the EQ-5D-5L.[11] This instrument differentiates between health problems on 5 domains: mobility, self-care, usual activities, pain and discomfort, and anxiety and depression, in that order. On each of those domains, one can have (1) no problems, (2) slight problems, (3) moderate problems, (4) severe problems, or (5) extreme problems. Thus, a health profile "15234," for example, indicates that a respondent has no problems with mobility, extreme problems with self-care, slight problems with their usual activities, moderate pain or discomfort, and severe anxiety and depression. In short, the order of the numbers indicate the domains and the number the respective level of severity on each domain.

Valuation Methods

Two valuation methods are compared in this experiment: the composite cTTO and the BTD method.[3,7] In the cTTO, utilities are determined by an indifference procedure in which respondents choose between living in good health for a certain amount of years or living in an inferior health state for another amount of years. Both of these health episodes start now and are followed by death. The indifference procedure starts at 10 years in good health versus 10 years in the health state of interest. If 10 years in good health is preferred, the respondent is asked to choose between 10 years in the health state or dying immediately. If the respondent prefers the 10 years in the health state, the respondent is asked to choose between 5 years in good health or 10 years in the health state. Depending on the subsequent responses, the time in good health is then increased or decreased in steps of 1 or 0.5 years until indifference is reached. If the respondent prefers dying immediately, another task is introduced. Now the 10 years in the health state is preceded by 10 years in good health, and the alternative is living in 10 years in good health followed by death. If the respondent prefers 10 years in good health, he or she is asked to choose between (5 years in good health) versus (10 years in good health, followed by 10 years in the health state). Again, the time in good health is then incremented or decremented in steps of 1 or 0.5 years until indifference is reached.[3] cTTO responses are considered to be cardinal.

The BTD method lets respondents choose between being in a health state for a certain number of years followed by death or dying immediately.[7] These health episodes start now and are followed by death. Responses are binary and ordinal and provide information on whether a respondent considers a health state of a certain duration as better or worse than dead. BTD valuation data can be analyzed using logistic models, such as random effects logistic regression.[7] These models for the binary BTD method data produce values on the 0 (dead) and 1 (full health) utility scale, in which the proportion of "dead" responses are used to infer a value for a health state. States that have a probability of 50% of being chosen over dead are then assigned the value 0, as the probability of picking dead or that health state is equal. This anchors the scale on dead equaling 0.

Table 1. Sample characteristics.

	n	%
Age, y		
18-24	111	55.5
25-34	62	31
35-44	13	6.5
45-54	4	2
55-64	10	5
65+	0	0
Sex		
Male	57	28.5
Female	143	71.5
Education		
Primary	1	0.5
Secondary	72	36
Tertiary	127	63.5
Problems on EQ-5D dimension		
No problems	123	61.5
Problems	77	38.5
Self-reported health		
Very good	109	54.5
Good	80	40
Fair	7	3.5
Fairly poor	3	1.5
Poor	1	0.5
Religious		
Yes	35	17.5
No	165	82.5
Believes in the afterlife		
Yes	47	23.5
No	153	76.5

Table 2. Response pairs to the BTD and cTTO tasks.*

		cTTO Method			
		WTD	ETD	BTD	Total (%)
BTDmethod	WTD	467	34	264	765 (54.7)
	BTD	54	14	566	634 (45.3)
	Total (%)	521 (37.2)	48 (3.4)	830 (59.3)	1399

BTD indicates better than dead; cTTO, composite time trade-off; ETD, equal to dead; WTD, worse than dead.
*The cTTO is truncated into the following responses: BTD (1), equal to dead (0), or worse than dead (−1). BTD responses are binary, either BTD (1) or WTD (−1).

Selection of Health States

To compare the 2 methods, health states need to be picked carefully. If the health states are too extreme, it is not possible to compare the 2 methods on consistency and test–retest reliability and to determine the agreement between the 2 methods, as nearly all responses will be in the same category. For example, for very mild states, nearly all respondents will choose better than dead in both methods. Health states are selected based on their vicinity to dead in the Dutch tariff, ensuring a wider spread in responses in both tasks.[12] In addition, dominance criteria were used for the selection, to maximize the number of possible inconsistencies. The included health states 34333, 23341, 32342, 44343, 45353, 33444, and 53345.

Interview

Respondents were interviewed in a face-to-face setting. Respondents started the experiment by filling in the EQ-5D-5L self-description form. After that, respondents completed either the paper-based BTD or the computer-based cTTO task, followed by a short questionnaire containing some sociodemographic questions. Next, the respondents completed the remaining cTTO task or BTD task, followed by a questionnaire containing remaining background questions. The retest of a single BTD question and a single cTTO question was performed at the end. After the interview, respondents were debriefed if desired. Each respondent completed the BTD task and the cTTO task, each task for the same 7 health states that were mentioned in the previous section.

Valuation Tasks

The computer-assisted cTTO task was done using the EuroQol-Portable Valuation Technology (EQ-PVT) software. For practice, respondents valued a wheelchair example, a mild practice state (12111), and the pits state (55555). Afterward, the same 7 health states were valued in a random order using the cTTO method.

Table 3. Agreement and polychoric correlations.

Health state	Agreement (%)	Polychoric correlation
34333	92.5	0.869
23341	87	0.789
32342	84	0.829
44343	65	0.572
33444	68	0.503
45353	69	0.553
53345	75	0.629
Total	77.1	0.791

Figure 1 in the appendix (in the Supplemental Materials found at https://doi.org/10.1016/j.jval.2019.10.006) shows an example question of an EQ-PVT cTTO question.

The paper-based BTD task started with a practice example in which respondents have to choose between being in a wheelchair for 10 years followed by death or dying immediately. Afterward, respondents were presented 7 choice pairs of (1) living in 1 of the 7 health states for 10 years, followed by death, or (2) dying immediately. The health states were divided into 4 sets with different orders of health states. Figure 2 in the appendix of the Supplemental Materials (found at https://doi.org/10.1016/j.jval.2019.10.006) shows an example of a BTD question as used in this experiment.

For the retest, half of the respondents valued the health state 44343 using the BTD method and health state 33444 using the cTTO method. The other half of the respondents valued health state 33444 using the BTD method and health state 44343 valued using the cTTO.

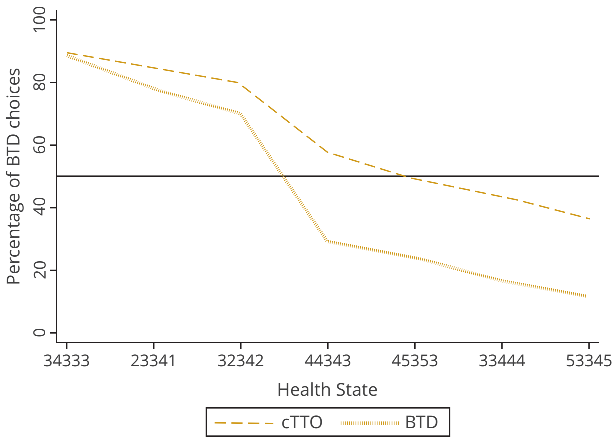
Quality Control

Quality control for the cTTO data collection was conducted after including every multiple of 20 respondents, as in Ramos-Goñi et al.[13] Following their protocol, interviews were flagged if 1 of the following criteria was met: the worse-than-dead task was not shown in the practice states, the overall explanation of the cTTO task took less than 3 minutes, or the respondent spent on average less than 30 seconds on each health state after the practice states. The last criterium in their protocol, that the state "55555" was valued at least 0.5 higher than another state, was not included, as "55555" was not in our valuation set.

Analyses

To determine an agreement measure between the cTTO and the BTD methods, responses to the second cTTO question were recoded. Responses were recoded from considered better than, equal to, or worse than dead to 1, 0, and −1, respectively. For the BTD task, responses were coded as 1 for better than dead and −1 for worse than dead. Agreement was analyzed in 2 ways: with or without considering equal-to-dead responses. In the first way, the agreement score between the cTTO and the BTD was defined to be either 1 or 0. Disagreement (0) occurred if the cTTO score was 1(−1) and the BTD score was −1(1). The agreement score was 1 otherwise. In the second way, agreement was set at 0 if a respondent chose equal to dead in the cTTO, whereas the rest of the computation stayed the same. As an additional measure of agreement, polychoric correlations were used. Polychoric correlations are suitable to estimate the correlation between 2 ordinal variables, such as Likert scale items, assuming that they measure latent variables that are normally distributed.[14] Because both methods measure the preference of a respondent for a health state, and we assume these preferences to be normally distributed

Figure 1. Percentage of better-than-dead responses for the cTTO and BTD method, by health state.



BTD indicates better than dead method; cTTO, composite time trade-off.

for each health state, this technique can be used to analyze both the ternary cTTO data and the binary BTD data. Polychoric correlations can be interpreted in the same way as the Pearson correlation coefficient. A measure of inconsistency was defined for both the cTTO and BTD method. A respondent would be inconsistent if he or she assigned a lower value to a logically better health state. In the cTTO task, 3 possible combinations were counted as inconsistent when state B dominates state A: (1,0), (1,−1) and (0,−1). For the BTD method, there is only 1 possible combination for inconsistency when state B dominates state A: (1,−1).

The set of health states allows for 12 possible dominant pairs of health states. For each of these 12 pairs of health states, inconsistencies are defined as either 0 (no inconsistency) or 1 (inconsistent). Thus, for each method, each respondent has an aggregate inconsistency score ranging between 0 (no inconsistencies) and 12 (all pairs are inconsistent).

To calculate the test–retest reliability for both methods, the agreement between the original answer and the retest was assessed. Agreement was now determined within each method. In the BTD method, answer pairs (1,1) and (−1,−1) would indicate agreement, whereas (−1,1) and (1,−1) would indicate disagreement. For the cTTO, (1,1), (−1,−1), and (0,0) would indicate agreement, and all other possibilities would indicate disagreement. In addition, polychoric correlations were used to quantify the test–retest reliability of the cTTO, whereas tetrachoric correlations were used to quantify the test–retest reliability of the BTD method. Tetrachoric correlations are a special case of polychoric correlations for which the outcomes of both variables are binary.[14]

Results

Two hundred respondents participated in the experiment, whose sample characteristics are reported in Table 1. One respondent missed 1 of the BTD questions, so 1399 BTD responses and 1400 cTTO responses were collected. The EQ-VT quality control protocol flagged 1 interview out of 200 to be suspicious, as the worse-than-dead task was not explained in the practice questions of the cTTO task. Given the very small number of flagged interviews, this interview was not excluded from analysis. The mean cTTO utilities did not depend on the task order of completing the cTTO or BTD task first, except for health state 33444 (t test, $P = .02$). The distribution of the responses by task is reported in Table 2.

Agreement scores were computed by health state and are reported in Table 3. In addition, polychoric correlations were used to quantify agreement, which is also reported in Table 3. Because there were few equal-to-dead responses, the 2 ways of calculating agreement did not differ. Agreement was higher for the health states that were assigned a relatively high (34333, 23341, and 32342) or a relatively low (53345) score in the Dutch EQ-5D-5L tariff compared with those that were closer to 0 (44343, 33444, and 45353). The polychoric correlations ranged from 0.865 for the best agreement to 0.503 for the lowest agreement. The polychoric correlation for the pooled data equaled 0.791. Chi-square tests were performed to assess order effects for the agreement between the cTTO and BTD and whether the cTTO or BTD task was completed first for each of the 7 health states. The agreement between the cTTO and BTD responses did not depend on the task order of completing the cTTO or BTD first for any of the health states.

Consistency scores and test–retest reliability were calculated for each respondent and method. Both methods scored high on consistency: for both tasks, about 91% of the respondents had no inconsistencies. In the cTTO method, 6% of the respondents had 1 inconsistency, 2.5% of the respondents had 2 inconsistencies, and 0.5% had 4 inconsistencies. In the BTD method, 7.5% had 1 inconsistency. In total, 35 respondents had at least 1 inconsistent response in either the cTTO or the BTD method, of which 1 respondent had at least 1 inconsistency in both the cTTO and the BTD method. Test–retest reliability was assessed, and both methods had a test–retest reliability score of 88%. The test–retest reliability was first calculated by health state, but values were pooled for the final analysis. Test–retest reliability was also assessed using tetrachoric correlation for the BTD method (0.8649) and polychoric correlation for the cTTO (0.897).

From Table 2, it can be seen that respondents chose WTD more often in the BTD task (54.7%) as compared with the cTTO task (37.2%). Equal-to-dead responses were rare in the cTTO (3.4% of cTTO responses). Figure 1 illustrates the amount of better-than-dead responses for each method, by valued health state; these states are ordered by their score on the Dutch EQ-5D-5L tariff.[12]

Additional variables of interest, such as the results from the task-difficulty questions, are reported in Table 4. There was no significant difference in reported “difficulty to understand the questions asked” nor in “difficulty to determine the difference between the valued health states” between the cTTO and BTD methods. The average time needed to answer a cTTO question was 74.3 seconds, whereas this was 26.5 seconds for a BTD question.

Discussion

Main Findings

The main findings of this study are that the overall agreement between the ternary cTTO and binary BTD method is high (77.2%). This agreement differed between the health states that were presented to the respondents and ranged between 92.5% and 65% for specific health states. Consistency scores were high for both methods, as was the test–retest reliability. Respondents valued health states more often as worse than dead in the BTD method.

Interpretation

The overall agreement was relatively high, suggesting that the cTTO and BTD method measure the same underlying preferences. Agreement was lower for health states close to dead, indicating that responses around dead differed more often between the 2 methods. Figure 1 illustrates this; the frequency of BTD responses

Table 4. Respondents scoring of task difficulty.

Easy to understand task	Very easy	Easy	Neutral	Hard	Very hard
cTTO	119	57	9	8	5
BTD	147	32	9	8	4
Seeing difference between health state	**Very easy**	**Easy**	**Neutral**	**Hard**	**Very hard**
cTTO	44	73	35	41	5
BTD	54	67	34	41	4

BTD indicates better than dead method; cTTO, composite time trade-off.

decreases when the health states become more severe for both methods. Lower agreement is expected when health states are close to dead because respondents may be unsure to value a state as better or worse than dead. This may be reflected in their responses, leading to more variation in responses for health states closer to dead and hence a lower agreement. Another contribution may be that, in the BTD method, more worse-than-dead responses were observed compared with the cTTO, which may also account for part of the disagreement between the 2 methods.

The amount of inconsistencies was low and about equal for both methods. This is reasonable, as we had a young, highly educated, mostly female sample, which are factors that have been shown to lead to fewer inconsistencies.[15,16] The amount of inconsistencies was comparable with those found in, for example, the Canadian EQ-5D-5L valuation study, in which 11% of respondents had at least 1 inconsistent response.[17,18]

The test–retest reliability was similar for both methods: 88% chose the same binary outcome in the BTD method as in the retest, and 88% chose the same ternary outcome in the second cTTO question. Other studies show varying amounts of test–retest reliability in (c)TTO studies.[1,19,20] Our study showed a test–retest reliability in cTTO data comparable with that of Purba et al[20] (95%), although different definitions of test–retest reliability were used.

Regarding task feasibility, the 2 tasks were judged to be of similar difficulty and similar ease of discriminating between health states, as shown by the responses to the task-difficulty questions. The duration of the tasks differed strongly: the time taken for each BTD question (26.5 seconds on average) was on average shorter than the 74.3 seconds spent on the cTTO. The BTD method also requires less explanation and supervision.

Why Are There More Worse-Than-Dead Responses in the BTD Method?

Health states were more often valued as worse than dead in the BTD method as compared with the cTTO method, which is illustrated in Figure 1. It is possible that response strategies cause the cTTO to have fewer worse-than-dead responses than the BTD method, and 3 of these response strategies are discussed. The first response strategy was put forward by the respondents themselves, indicating that "they were wavering, but if a choice had to be made between immediate death and living in a severe health state for 10 years, the cTTO task gave them the additional possibility to trade a lot of years on the positive board, instead of going to the negative board." In the BTD method, however, respondents were constricted to pick a worse-than-dead response for such a severe health state. A second response strategy is that cTTO respondents may wish to avoid the WTD task and therefore more often choose "better than dead" in the cTTO, as respondents often find the cTTO worse-than-dead task difficult and confusing.[21] If this is true, this may reduce the validity of cTTO responses of those respondents. A third response strategy is already known in cTTO valuations, in which respondents switch back from the WTD task to the BTD task. A reason for this may be that in the WTD task, the lead time approach is used in which 10 years in good health is added before the health state to be judged, which may change the attractiveness of the health state.[22] The switch back was not observed in our study: out of 1400 valuations, only 2 switches from the worse than dead back to the BTD task were made. All in all, the aforementioned response strategies led to more BTD responses in the cTTO method.

Besides the response strategies mentioned earlier, other explanations may be possible as well, such as differences in the framing of the tasks between the 2 methods, the fact that the BTD or worse-than-dead question is embedded in an indifference procedure in the cTTO, or the availability of the "equal to dead" possibility in the cTTO. In general, it is striking that although the questions in the valuation tasks and the measurement properties of the 2 methods are similar, worse-than-dead responses were more common in the BTD method.

Limitations and Strengths

A limitation of this study is that we used a highly educated sample. A more representative sample might have led to a greater amount of inconsistencies and possibly lower agreement. Another limitation is that more health states could have been included per respondent. However, because there is a delicate balance between burdening respondents and obtaining more data per respondents, and because we compared 2 methods, we chose to include only 7 states, so as not to burden the respondents. Lastly, a limitation of this study is that we could have measured maximum endurable time preferences and tested the constant proportional trade-off assumption and compared these between the 2 methods. However, this would have required more time frames to be included, increasing the burden for the respondents.

A strength of this study is that it is the first study to compare 2 different valuation methods, the cTTO and the BTD, in the same respondents. Second, all the interviews were done face to face and by the same interviewer, avoiding interviewer effects. Third, the EQ-VT protocol was followed, including the quality control process. Lastly, the choice of health states was appropriate, as several health states were in the vicinity of death.

Conclusion

The agreement between the 2 methods suggests that they measure the same underlying preferences. Furthermore, the measurement properties of the 2 methods were similar, as both methods had similar rates of inconsistency and test–retest reliability. It was observed that respondents chose worse than dead more often in the BTD method compared with the cTTO. Thus, compared with the cTTO, the BTD method favors life-improving values instead of life-saving values.[23] More research is needed to explore the measurement properties of the BTD method in the valuation of health states, as the BTD method is a simple and fast valuation method and is able to measure maximum endurable time preferences.[7]

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Supplemental Material

Supplementary data associated with this article can be found in the online version at https://doi.org/10.1016/j.jval.2019.10.006.

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